Groundtrack of GPS constellation and LEO satellites

Oscar Gonzalo Aran
oscar.gonzalo@estudiantat.upc.edu

Adam Ruiz Herrero
adam.ruiz@estudiantat.upc.edu

***Abstract*—This report describes how the ground tracks of two sets of satellites are computed and plotted with MATLAB: (i) the whole GPS constellation, from the YUMA almanac published by CelesTrak, over the next 60 minutes, and (ii) the Iridium NEXT satellite IRIDIUM 170 (Group 18), from its two-line elements (TLE), over one complete orbit. Both cases share the same chain of computations: Keplerian propagation of the orbit, rotation from the orbital plane to the Earth-fixed frame, and a closed-form conversion from ECEF to WGS-84 geodetic coordinates, programmed by the authors. The GPS result is compared with the real-time satellite-status map of the FAA and the Iridium result with the reference plot supplied for the group. In both cases the results agree with the references. At 20:25:08 UTC on 30/09/2026 the sub-satellite point of IRIDIUM 170 is found at latitude -22.015°, longitude -7.867° and altitude 657.3 km.**

***Index Terms*—ground track, GPS almanac, YUMA, two-line elements, Iridium, Kepler equation, ECEF, WGS-84.**

I. Introduction

The ground track of a satellite is the locus of its sub-satellite points, i.e. the points of the Earth's surface that lie on the line joining the satellite and the centre of the Earth. This second session extends the single-satellite program of Lab. Session 1 to two new situations, which set the two objectives of the session [1]:

1) Plot the ground track of *all* the satellites of the GPS constellation for a time lapse of one hour, starting at the moment the script is run ("NOW"), using the almanac of CelesTrak [2];
2) Learn the TLE format and use it to plot the ground track of one Iridium NEXT satellite during a complete orbit.

Our group (Group 18) was assigned the satellite IRIDIUM 170 (NORAD 43930). The code is organised in the main scripts `main2.m` (GPS constellation) and `main3.m` (Iridium), and in the functions `mean_anomaly`, `eccentric`, `true_anomaly`, `orbitcoords`, `satecef` and `ecef2lla`, which are called by `subsatellite.m`. Two helper functions supplied with the lab, `almanac.m` and `time2toa.m`, provide the almanac data and the time offset from the epoch of the elements. The remainder of the report follows the order of the lab statement: Section II presents the propagation model that both programs share, Section III the full GPS constellation, Section IV the TLE format and the Iridium program, Section V the ECEF-to-LLA algorithm, and Section VI the results.

II. Propagation model

The constants used in all programs are the gravitational constant $G = 6.67384 \cdot 10^{-11}$ m^{3}kg^{-1}s^{-2}, the Earth mass $M_e = 5.972 \cdot 10^{24}$ kg, the Earth rotation rate $\omega_e = 7.2921151467 \cdot 10^{-5}$ rad/s and the convergence tolerance for the Kepler equation, 10^{-8}.

A. Time reference

The user gives a UTC date (or the current time is read from the system). The Julian Date is computed with the algorithm of `JDfromUTC.m`,

$$JD = \lfloor 365.25\,y \rfloor + \lfloor 30.6001\,(m+1) \rfloor + D + \frac{h + \min/60 + s/3600}{24} + 1720981.5, \quad (1)$$

with $(y, m) = (Y, M)$ if $M > 2$ and $(y, m) = (Y-1, M+12)$ otherwise. GPS time differs from UTC by the accumulated leap seconds (18 s at the time of the lab). The GPS time in days since the GPS epoch ($JD_0 = 2444244.5$, i.e. 6 January 1980) is

$$t_{GPS} = JD - JD_0 + \frac{18}{86400}, \quad (2)$$

the GPS week is $w = \lfloor t_{GPS}/7 \rfloor$ (broadcast modulo 1024) and the number of seconds elapsed since the week epoch (Sunday 00:00 GPS time) is

$$e_{sec} = 86400\,(t_{GPS} - 7w). \quad (3)$$

B. From orbital elements to the sub-satellite point

Given the Keplerian elements of a satellite at its epoch t_{oa} (eccentricity e, inclination i, semi-major axis a, argument of perigee ω, mean anomaly M_0, and the longitude Ω_0 of the ascending node, with its drift $\dot{\Omega}$), the sub-satellite point at time t is obtained as follows (Fig. 1); the same chain is executed by `subsatellite.m` at every time step, and every intermediate result is written to the debugging file.

Step 1 – elapsed time and node longitude:

$$\Delta t = t - t_{oa}, \qquad \Omega_0 = \Omega_{week} - \omega_e\, t_{oa} \quad \text{(almanac)}. \quad (4)$$

Step 2 – mean anomaly: With the mean motion $n = \sqrt{GM_e/a^3}$ (rad/s),

$$M = M_0 + n\,\Delta t. \quad (5)$$

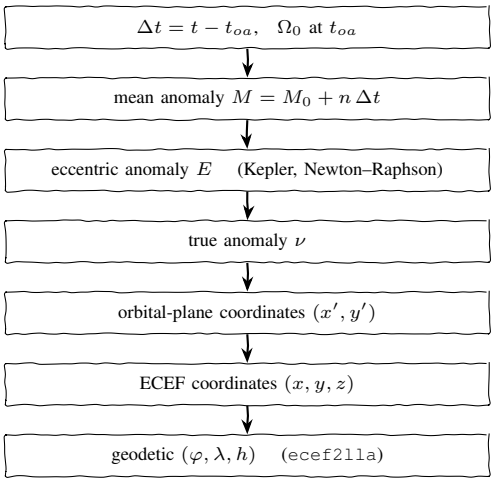


Fig. 1. Computation chain implemented in `subsatellite.m`.

Step 3 – eccentric anomaly: Kepler's equation $E - e\sin E = M$ has no closed-form solution and is solved with Newton–Raphson iterations starting at $E_0 = M$,

$$E_{k+1} = E_k - \frac{E_k - e\sin E_k - M}{1 - e\cos E_k}, \tag{6}$$

until $|E_{k+1} - E_k| < 10^{-8}$ (function `eccentric.m`).

Step 4 – true anomaly:

$$\nu = 2\,\mathrm{atan2}\big(\sqrt{1+e}\,\sin\tfrac{E}{2},\ \sqrt{1-e}\,\cos\tfrac{E}{2}\big)\,. \tag{7}$$

The `atan2` form avoids any quadrant ambiguity.

Step 5 – coordinates in the orbital plane: With the orbital radius $r = a(1 - e\cos E)$ and the argument of latitude $u = \omega + \nu$,

$$x' = r\cos u, \qquad y' = r\sin u. \tag{8}$$

Step 6 – ECEF coordinates: The ascending node, measured in the Earth-fixed frame, moves with the drift $\dot\Omega$ of the orbital plane and against the rotation of the Earth,

$$\Omega(\Delta t) = \Omega_0 + (\dot\Omega - \omega_e)\,\Delta t, \tag{9}$$

and the rotation from the orbital plane to the ECEF frame (function `satecef.m`) is

$$x = x'\cos\Omega - y'\cos i\,\sin\Omega, \tag{10}$$

$$y = x'\sin\Omega + y'\cos i\,\cos\Omega, \tag{11}$$

$$z = y'\sin i. \tag{12}$$

Step 7 – geodetic coordinates: The ECEF position is converted to latitude, longitude and altitude over the WGS-84 ellipsoid with the algorithm of Section V.

Step 8 – ground track: The ground track is obtained by repeating Steps 2–7 for $\Delta t_k = \Delta t_0 + k\,\delta t$ and plotting (λ_k, φ_k) over the world map drawn by `plotworld.m`.

TABLE I
COLUMNS OF THE MATRIX EPH (YUMA ALMANAC).

Col.	Field	Symbol	Unit
1	ID (PRN)	`id`	–
2	Health (0 = usable)	–	–
3	Eccentricity	e	–
4	Time of applicability	t_{oa}	s of week
5	Orbital inclination	i	rad
6	Rate of right ascension	$\dot\Omega$	rad/s
7	$\sqrt{a}$	$a = \mathrm{Eph}(s,7)^2$	$\mathrm{m}^{1/2}$
8	Right ascension at week	Ω_{week}	rad
9	Argument of perigee	ω	rad
10	Mean anomaly at t_{oa}	M_0	rad
11, 12	Clock bias, drift	a_{f0}, a_{f1}	s, s/s
13	GPS week (mod 1024)	–	–

III. GROUND TRACK OF THE FULL GPS CONSTELLATION

A. Adaptation of the Lab. Session 1 program to the next hour

The program of Lab. Session 1 (`main1.m`), which tracked one satellite for 24 h with hard-coded ephemeris, was renamed `main2.m` and its debugging file changed to `main2.dat`. The time loop was changed to span from NOW to NOW plus 60 minutes, with a step of 60 s, i.e. $k = 0, \ldots, 60$, which gives 61 positions per satellite. To compute the position for NOW and compare it with the FAA, the almanac must be the one valid at the time the script is run; therefore the hard-coded ephemeris were replaced by the ones read from the almanac, as explained next.

B. The almanac and the matrix *Eph*

The GPS almanac in YUMA format contains, for each satellite, the 13 fields of Table I. The function `almanac.m` downloads the most adequate file from CelesTrak (if it is not already in the working folder) and returns the current GPS week `baseweek`, the seconds from the week epoch `esec`, the number of healthy satellites `NS` and the matrix `Eph` of size $NS \times 13$, one satellite per row. It is called **once** in `main2.m`:

```
[baseweek, esec, NS, Eph] = almanac();   % NOW
```

Called with arguments, `almanac(DD,MM,YYYY,hh,mm,ss)` "freezes" the time at that UTC instant, which is used to reproduce past situations.

For satellite s (row s of `Eph`) the variables of the propagation model are assigned as

```
id = Eph(s,1);  e  = Eph(s,3);  toa = Eph(s,4);
io = Eph(s,5);  dO = Eph(s,6);  a   = Eph(s,7)^2;
Lo = Eph(s,8);  w  = Eph(s,9);  Mo  = Eph(s,10);
dt0 = esec - toa;     Oo = Lo - Oe*toa;
```

and for $k = 0, \ldots, 60$ the function `subsatellite` is called with $\Delta t = \Delta t_0 + 60\,k$. The clock terms a_{f0}, a_{f1} are not used. Each satellite is plotted with its own colour (`hsv(NS)`); the position for NOW ($k = 0$) is marked with a circle and the PRN number.

C. Selection of the almanac

The almanacs are generated at the following times of applicability (seconds of the GPS week):

$$t_{oa} \in \{61440,\ 147456,\ 233472,\ 319488,\ 405504,\ 503808,\ 589824\}.$$

`almanac.m` chooses the first t_{oa} in this list for which

$$e_{sec} + 3600 < t_{oa} + 7200, \tag{13}$$

i.e. the almanac must still be valid (obsolescence limit of 2 h) during the whole hour that is going to be plotted. If no such t_{oa} exists in the current week, the first ones of the next week are used and e_{sec} becomes negative. As a consequence, $\Delta t_0 = e_{sec} - t_{oa}$ may be negative, which only means that the almanac epoch lies after the instant of interest; the equations of Section II are valid for either sign of Δt.

IV. Two-line elements and the Iridium program

A. TLE format

The TLE is the standard format for the Keplerian elements of most civilian satellites [3]. The TLE of our satellite, supplied in `Group_18.tle`, is

```
IRIDIUM 170
1 43930U 19002J   26273.85494888 .00000304 00000+0 44338-4
     0  9997
2 43930  86.6070 309.3776 0002386  91.2600 268.8890
     14.72247722414806
```

Only the following fields are used. From the first line, the fourth field gives the epoch: "26" is the last two digits of the year (2026) and 273.85494888 is the time of applicability in days from 00:00 UTC of 1 January, i.e. 30/09/2026 at 20:31:07.6 UTC. From the second line, in the order in which they appear: inclination $i = 86.6070°$, right ascension of the ascending node $\alpha_\Omega = 309.3776°$, eccentricity $e = 0.0002386$ (the field, 0002386, is $e \cdot 10^7$), argument of perigee $\omega = 91.2600°$, mean anomaly $M_0 = 268.8890°$ and mean motion $N = 14.72247722$ rev/day. The last six digits (414806: revolution number and checksum) are ignored.

B. Time offset and GAST

The function `time2toa(toa, DD, MM, YYYY, hh, mm, ss)` returns the elapsed time Δt from the epoch of the elements to the given UTC date, and the Greenwich Apparent Sidereal Time (GAST) at the epoch, in degrees. Δt is computed from Julian Dates,

$$\Delta t = 86400\,\big(JD - (JD_{1/1/Y} + t_{oa} - 1)\big), \tag{14}$$

where t_{oa} is the epoch in days of the year. The GAST is the right ascension of the Greenwich meridian and it is what allows right ascension angles to be converted into longitudes. It is obtained from the Earth rotation angle (ERA) and a polynomial in Julian centuries T (function `greenwichMeanSiderealTime.m`):

$$\begin{aligned}\mathrm{ERA} = 2\pi\big(&0.7790572732640 + \mathrm{frac}(JD) \\ &+ 0.00273781191135448\,(JD - 2451545)\big),\end{aligned} \tag{15}$$

$$\begin{aligned}\mathrm{GAST} = \mathrm{ERA} + \frac{\pi}{180\cdot 3600}\big(&0.014506 \\ &+ 4612.15739966\,T + 1.39667721\,T^2 \\ &- 0.00009344\,T^3 + 0.00001882\,T^4\big).\end{aligned} \tag{16}$$

(The implementation neglects the nutation correction, so in practice it returns the mean sidereal time.)

C. Conversion of the TLE data (Step 2)

The TLE does not contain the semi-major axis a, the longitude of the ascending node Ω_0 at t_{oa} or the nodal drift $\dot\Omega$, which `subsatellite.m` needs, and its angles are in degrees. The following conversions are made:

$$n = \frac{2\pi N}{86400}\ \text{[rad/s]}, \qquad a = \left(\frac{GM_e}{n^2}\right)^{1/3}, \tag{17}$$

$$\Omega_0 = \alpha_\Omega - \mathrm{GAST}(t_{oa}) \quad \text{[rad]}, \tag{18}$$

$$i,\ \omega,\ M_0 \text{ from degrees to radians}, \qquad \dot\Omega = 0. \tag{19}$$

Eq. (17) is Kepler's third law. Eq. (18) turns the inertial right ascension of the node into a longitude, i.e. an angle measured from the Greenwich meridian, which is the angle that the model of Section II uses. The drift is set to zero because the TLE contains no information about it.

D. Program structure

`main3.m` follows the skeleton of the statement literally. The key lines are:

```
[dt, GASTdegToA] = time2toa(toa, DD,MM,YYYY,hh,mm,ss);
n_rad = n*2*pi/86400;
a  = (G*Me/n_rad^2)^(1/3);
Oo = deg2rad(raan - GASTdegToA);
io = deg2rad(io); w = deg2rad(w);
Mo = deg2rad(Mo); dO = 0;
fprintf(fd, "io = %f, a = %f, ...", ...);  % debug
[long,lat] = subsatellite(Oe,G,Me,a,Mo,dt,e,tol,w,Oo,dO,io)
    ;
plot(long, lat, 'r*');
T = 2*pi/n_rad;  paso = 30;  N = ceil(T/paso);
for k = 1:N, [long,lat] = subsatellite(..., dt+k*paso, ...)
    ;
            plot(long, lat, 'r.'); end
```

The date and time given in the file `Group_18.pdf` (and in the title of the reference plot) are 30/09/2026 at 20:25:08 UTC. Once the position at that instant has been plotted (Step 3), the time is increased in steps of 30 s during one orbital period $T = 2\pi/n = 5868.6$ s (Step 4). This requires $\lceil T/30 \rceil = 196$ steps.

V. Conversion from ECEF to LLA

Although MATLAB has a native function, the conversion was programmed by the authors (`ecef2lla.m`) with the closed-form, non-iterative algorithm of the statement [1]. The input is (x, y, z) in metres and the datum is WGS-84: $a = 6378137$ m, $e^2 = 0.00669437999014$, $b = a\sqrt{1-e^2}$ and

$e'^2 = (a^2 - b^2)/b^2$. (Here a, b are the ellipsoid parameters and have nothing to do with the orbital a.) The algorithm is

$$r = \sqrt{x^2 + y^2}, \qquad E^2 = a^2 - b^2, \tag{20}$$

$$F = 54\, b^2 z^2, \qquad G = r^2 + (1 - e^2) z^2 - e^2 E^2, \tag{21}$$

$$C = \frac{e^4 F r^2}{G^3}, \qquad s = \sqrt[3]{1 + C + \sqrt{C^2 + 2C}}, \tag{22}$$

$$P = \frac{F}{3\left(s + \frac{1}{s} + 1\right)^2 G^2}, \qquad Q = \sqrt{1 + 2e^4 P}, \tag{23}$$

$$r_0 = -\frac{P e^2 r}{1 + Q} + \sqrt{\tfrac{1}{2} a^2 \Big(1 + \tfrac{1}{Q}\Big) - \frac{P(1 - e^2) z^2}{Q(1 + Q)} - \tfrac{1}{2} P r^2}, \tag{24}$$

$$U = \sqrt{(r - e^2 r_0)^2 + z^2}, \tag{25}$$

$$V = \sqrt{(r - e^2 r_0)^2 + (1 - e^2) z^2}, \qquad z_0 = \frac{b^2 z}{a V}, \tag{26}$$

and the output is

$$h = U \left(1 - \frac{b^2}{a V}\right), \tag{27}$$

$$\varphi = \operatorname{atan2}\big(z + e'^2 z_0,\ r\big), \qquad \lambda = \operatorname{atan2}(y, x). \tag{28}$$

The function `atan2` gives the correct quadrant. The angles are returned in degrees and the output vector is `lla = [lat, long, alt]`. Since the function has the same name and format as the native one, MATLAB uses ours instead of the native one as soon as it is in the working folder. To compare both, it is enough to rename our function or to move it out of the path. As an independent check, our implementation was compared with the conversion based on an auxiliary angle (Bowring's formula) used in Lab. Session 1, over 2000 random points with geocentric distances between 6400 and 36400 km: the maximum differences were $4.8 \cdot 10^{-7}$ degrees in latitude, 0 in longitude and 0.29 m in altitude, which are negligible for plotting purposes.

VI. Results

A. Full GPS constellation

Two runs of `main2.m` are presented (Table II). The first one is the run for NOW, as required by the statement. The second one uses a frozen time that coincides with the time stamp of the FAA map, because the FAA tool shows the *last* hour, so both maps can only be compared at the instant given by the time stamp of its capture.

Fig. 2 shows the result for NOW. All the tracks are short arcs, which is expected: the orbital period of GPS is $T = 2\pi\sqrt{a^3/GM_e} = 11.97$ h for $a \approx 26559$ km, so in one hour a satellite covers about $1/12$ of its orbit ($\approx 30^\circ$ of argument of latitude). No track goes beyond about $\pm 55^\circ$ of latitude, which is the inclination of the GPS orbital planes ($i \approx 0.96$ rad). The satellites move eastwards with respect to the ground, because their angular rate (about 2 rev/day) is roughly twice the rotation rate of the Earth. The altitude returned by `ecef2lla` for PRN-01 at the initial instant is 20135.7 km, consistent with the nominal GPS altitude of about 20200 km.

TABLE II
Parameters of the two GPS runs.

	Run for NOW	Frozen time (FAA)
UTC date	08/10/2026 13:23:31	05/10/2026 20:45:07
GPS week	391 (2439)	391 (2439)
e_{sec} [s]	393828.58	161125.00
Almanac file	`0391.405504`	`0391.233472`
Δt_0 [s]	-11675.4	-72347.0
Satellites NS	32	32
Positions plotted	32×61	32×61

TABLE III
Position of some satellites at 20:45:07 UTC (05/10/2026): our program and graphical reading of the FAA map (degrees).

PRN	Computed (λ, φ)		FAA, read (λ, φ)	
18	15.4	20.1	≈ 17	≈ 20
04	-104.5	46.8	≈ -100	≈ 47
09	-150.3	55.5	≈ -147	≈ 56
26	-38.7	53.2	≈ -36	≈ 53
22	165.4	-54.4	≈ 166	≈ -53
06	166.3	21.4	≈ 167	≈ 22

Fig. 3 compares our result with the FAA map. In the FAA map the squares mark the current position of every PRN and the symbols behind them (one per orbital plane, A–F) the position during the last hour; in our map the circles mark the same instant and the arcs go forward in time. The comparison therefore has to be made between squares and circles, and the arcs must leave each marker in opposite senses (backwards in time in the FAA map and forwards in ours), which is what is observed. Table III lists the position computed by our program for six satellites together with the position read graphically from the FAA map (the reading error is a few degrees, since the squares are about 7° wide); all of them agree within that error. The 32 satellites of the almanac are present in both maps.

B. IRIDIUM 170

Table IV gathers the input data and the intermediate variables written to `main3.dat`. The satellite flies at an altitude of 657.3 km above the WGS-84 ellipsoid, which is below the nominal operational altitude of Iridium NEXT (about 780 km); this is consistent with the mean motion of this TLE ($N = 14.72$ rev/day), which is larger than that of the example TLE of the statement (14.34 rev/day).

Fig. 4 compares the result of `main3.m`, computed with our own `ecef2lla.m`, with the reference plot: the initial position ($\lambda \approx -7.9^\circ$, $\varphi \approx -22^\circ$), the northernmost and southernmost latitudes, the shape and the longitudes of the two branches coincide. The track is almost a meridian line because the inclination ($i = 86.6^\circ$) is close to 90°; the track goes north along $\lambda \approx -8^\circ$, passes close to the pole and comes down along $\lambda \approx 159^\circ$ on the other side of the Earth. The track does not close after one orbit: during one period the Earth

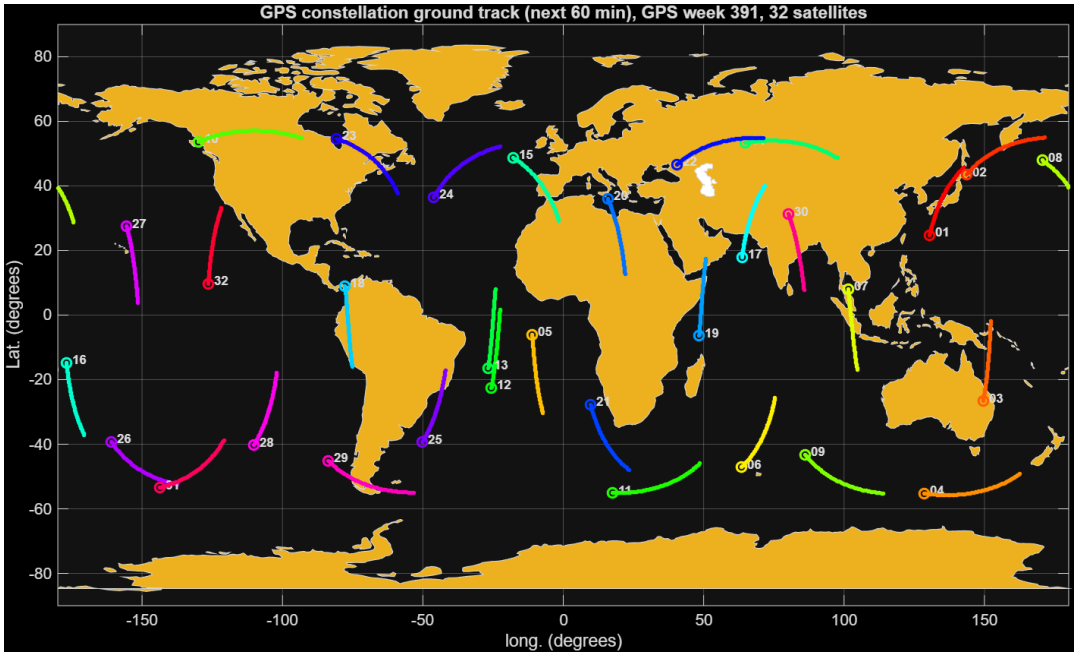


Fig. 2. Ground track of the full GPS constellation for the next 60 minutes starting at 13:23:31 UTC of 08/10/2026 (GPS week 391, 32 satellites). The circle and the number mark the position of each PRN at the initial instant; one point is plotted every 60 s.

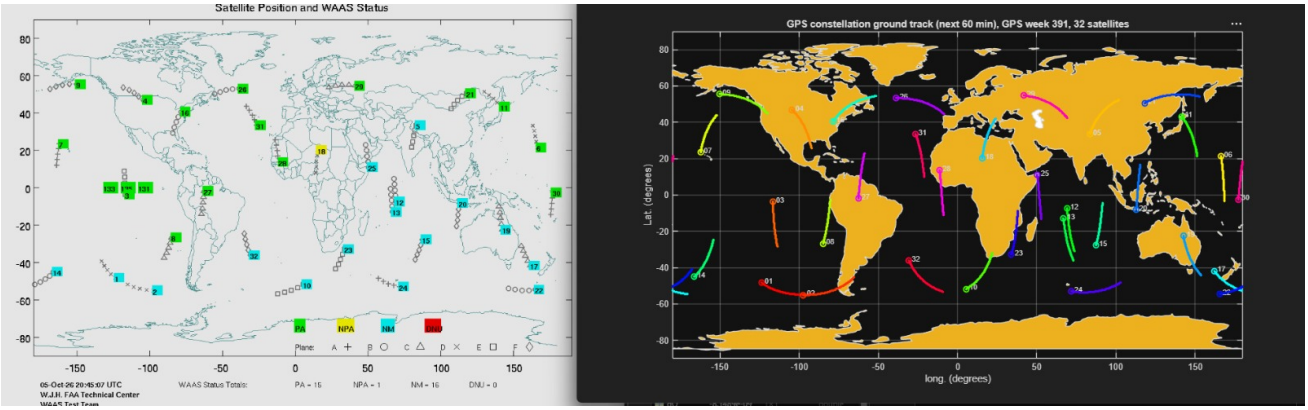


Fig. 3. Comparison with the FAA tool [4]. Left: capture of the FAA "Satellite Position and WAAS Status" map, time stamp 05-Oct-26 20:45:07 UTC (it shows the last hour; the coloured squares mark the current position). Right: our result for the same instant (frozen time), where the circles mark the position at 20:45:07 UTC and the arcs show the following hour.

rotates $\omega_e T = 24.5°$, so the satellite comes back to the same latitude at a longitude $24.5°$ to the west of the initial one, $-7.9° - 24.5° \approx -32.4°$, which coincides with where the track ends in the plot. Setting $\dot{\Omega} = 0$ has a negligible effect in this time lapse: the nodal regression due to J_2 at this altitude and inclination is about $-8.5 \cdot 10^{-8}$ rad/s, i.e. only $0.03°$ per orbit.

C. Debugging files

The files `main2.dat` and `main3.dat` contain, for every call of the functions, the intermediate variables of the chain of Fig. 1 (Δt, M, E, ν, (x', y'), Ω, ECEF coordinates and LLA). The first lines of `main3.dat` give the values of Table IV. The prototype of `ecef2lla` given in the statement prints the labels "lat" and "long" with the values in the opposite order; in our implementation each value is printed under its own label.

VII. Conclusions

- The same Keplerian propagation chain (Steps 1–8) and the same ECEF-to-LLA conversion were used for the whole GPS constellation (almanac data) and for an Iridium NEXT satellite (TLE data). The only differences are in how the inputs are obtained: for the almanac the semi-major axis is $(\sqrt{a})^2$, the longitude of the node is corrected with $-\omega_e t_{oa}$ and the nodal drift is broadcast; for the TLE, a follows from Kepler's third law, the node longitude is $\alpha_\Omega - \mathrm{GAST}(t_{oa})$ and $\dot{\Omega} = 0$.

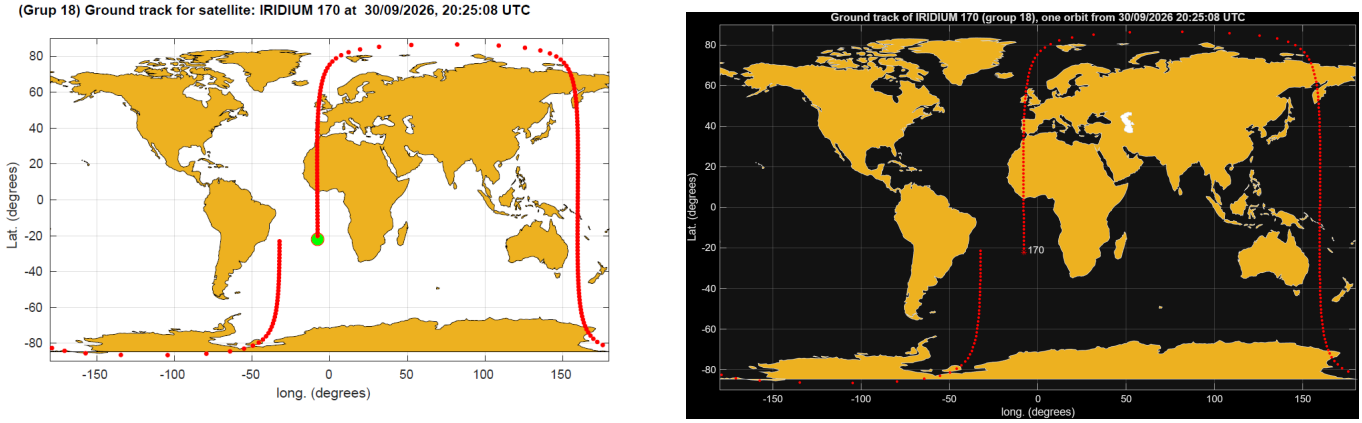


Fig. 4. Ground track of IRIDIUM 170 during one orbit starting at 20:25:08 UTC of 30/09/2026. Left: reference plot supplied for Group 18. Right: result of `main3.m`, computed with our own `ecef2lla.m`; the star marks the position at the initial instant and one point is plotted every 30 s.

TABLE IV
IRIDIUM 170 – VALUES IN `MAIN3.DAT` FOR THE INSTANT 30/09/2026 20:25:08 UTC.

Quantity	Value
Δt from epoch (20:31:07.6)	-359.583 s
GAST at epoch	$317.381°$
$\Omega_0 = \alpha_\Omega - \mathrm{GAST}$	6.143496 rad $(351.996°)$
Mean motion n	$1.070649 \cdot 10^{-3}$ rad/s
Semi-major axis a	7031.811 km
Period T	5868.58 s $(97.81$ min$)$
$i,\ \omega,\ M_0$	$1.511577,\ 1.592787,\ 4.692998$ rad
$M,\ E,\ \nu$ at the instant	$4.308011,\ 4.307792,\ 4.307572$ rad
(x', y')	$(6523.411, -2626.930)$ km
ECEF (x, y, z)	$(6463.858, -893.085, -2622.325)$ km
Latitude, longitude, altitude	$-22.015°,\ -7.867°,\ 657.3$ km

- The positions computed for the 32 GPS satellites agree with the FAA real-time map at the same instant, and the ground track of IRIDIUM 170 reproduces the reference plot supplied for the group.
- The closed-form ECEF-to-LLA algorithm programmed by the authors agrees with the conversion used in the previous lab to within $5 \cdot 10^{-7}$ degrees and 0.3 m.
- The model is a two-body Keplerian one: the effect of J_2 is only included through the almanac drift $\dot{\Omega}$ (GPS) and it is neglected for the TLE, which is acceptable for plotting a ground track over a few minutes or one orbit. A higher accuracy (for instance to predict passes) would require a perturbed propagator such as SGP4.
- The execution of the program for NOW depends on the almanac being available in CelesTrak. The results of this report were obtained with the almanac files included in the delivered archive, so they can be reproduced with the frozen-time call of `almanac`.

REFERENCES

[1] J. Olmos, "Lab. Session 2: Ground track of full GPS constellation and LEO satellites," Lab statement, Navegació Aèria, Cartografia i Cosmografia, EETAC-UPC, 2025–26.

[2] CelesTrak, "GPS Yuma almanacs." [Online]. Available: https://celestrak.org/GPS/almanac/Yuma/

[3] CelesTrak, "Satellite Times" column, vol. 4, no. 3 (TLE format). [Online]. Available: http://celestrak.com/columns/v04n03/

[4] FAA William J. Hughes Technical Center, "WAAS satellite status." [Online]. Available: https://www.nstb.tc.faa.gov/rt_waassatellitestatus.htm